



Five Stars, Same Leak: Satisfaction-Prompt Timing, Response Composition, and Recorded Repeat Callbacks Across a Home-Repair Platform's Staggered Fourteen-Day Rollout

Mirela Hanke, Marit Osayande, Dagny Okafor

School of Continuous Improvement, Slop University

{mirela.hanke, marit.osayande, dagny.okafor}@slop.university

doi:10.5555/slop.6fso51

Abstract. Home-repair booking platforms collect a satisfaction rating at the moment a tradesperson marks a job complete, and rank tradespeople on the resulting mean. We use one national platform's staggered, region-by-region migration of that prompt from immediate dispatch to a fourteen-day delay — six waves across nine metropolitan regions over 22 weeks, covering 38,714 completed jobs — to ask what the earlier prompt is in a position to measure. Ratings fall under the delayed prompt: a difference-in-differences estimate of -0.55 points on the five-point scale (95% CI -0.63 to -0.47), with the response rate falling from 68.4% to 41.2% and the one-star share rising from 1.7% to 10.0%. The platform's own recorded outcome does not move. Repeat callbacks to the same address for the same fault category within ninety days sit at 11.3% before the switch and 11.5% after, a difference-in-differences of $+0.2$ percentage points (95% CI -0.9 to $+1.3$) that holds across six specifications. Tradesperson-level ratings collected immediately are uncorrelated with that tradesperson's callback rate ($r = 0.04$); ratings collected at fourteen days are modestly negatively associated ($r = -0.31$). We observe that prompt timing appears to determine the informational content of the score rather than the durability of the work, and that the ranking surface has to date been fitted to the uninformative variant.

1. Introduction

The van pulls away and the phone buzzes. On most home-repair booking platforms the two events are separated by whatever time it takes a tradesperson to press a completion button on a handset, and the rating a household submits in that window becomes a permanent input to the platform's ranking of who gets offered the next job on that street.

The arrangement has an obvious convenience: response rates are high while the visit is fresh, and the record closes the same day. It also has a property that is less often stated, which is that the household is being asked to assess a repair before the repair has had an opportunity to fail. A tap that is dry at the moment the toolbox

goes back in the van is not yet a tap that holds. The score is collected at the one point in the job's life at which the question it asks cannot be answered.

We are able to observe what that costs because one national platform changed it. Over 22 weeks the platform migrated its post-job satisfaction prompt from immediate dispatch to a fourteen-day delay, region by region, in six waves — a staggered adoption that supports a difference-in-differences design against the platform's own operational records. Our contributions are as follows.

- A difference-in-differences estimate of the effect of prompt timing on submitted ratings and on response composition, which suggests the

delay changes what is submitted rather than who is competent.

- A parallel estimate on the platform’s recorded ninety-day repeat-callback rate, which we report as a null and which appears stable across the six specifications we examined.
- A comparison of the association between tradesperson-level ratings and recorded callbacks under the two timings, which may indicate that only the delayed instrument carries outcome information.

We take no position on which prompt a platform ought to use. We observe only that the two prompts measure different things, and that the platform’s ranking surface was not built on the assumption that they might.

2. Background

Customer satisfaction has been treated as an artefact of its own measurement for as long as it has been measured routinely [1], and the gap between a satisfaction instrument and a service-quality one was mapped early [2], [3]. The retrospective-evaluation literature is more pointed still: what a respondent reports about an episode depends heavily on when they are asked and on which moments of it are available to them [4], [5], and survey answers about one’s own recent history are reconstructions rather than readings [6].

The dissociation between satisfaction and outcome is documented most sharply in health services, where higher patient satisfaction has been associated with greater utilisation and expenditure [7], where the relationship between experience measures and clinical outcomes has required careful separation [8], and where the concept itself has been reviewed as under-specified [9]. Platform reputation systems inherit the problem in a compressed form: online feedback mechanisms carry known selection and inflation pressures [10], [11], ratings distributions on large marketplaces sit implausibly close to their ceiling [12], non-reporting is informative rather than neutral [13], and displayed scores influence subsequent ones [14]. Single-number summaries built on top of such instruments have not always predicted the outcomes claimed for them [15]. Response rate is not itself the binding constraint on bias [16], a point our own results bear on.

Prior work at this institution has twice found an instrument’s timing to outrank its content. A restroom satisfaction kiosk’s button-press timestamps were matched against a cleaning contractor’s service-scan log, and the highest ratings clustered tightly against the nearest preceding scan while every other rating spread evenly between rounds [17]. An internal-transfer portal’s status field was found to freeze at “Under review” while review activity continued behind it, with applicants shown the frozen field checking back for a median of forty-six days longer than applicants shown nothing at all [18]. The present study asks the same question of an instrument whose output is not merely displayed but ranked on.

3. Method

3.1. Setting and rollout

The platform brokers domestic repair work in five trades — plumbing, electrical, appliance repair, locksmithing, and heating and cooling — across nine metropolitan regions. Before the change, a satisfaction prompt was dispatched to the booking household within four minutes of the tradesperson marking the job complete. After the change, the same prompt, with identical wording and an identical five-point scale, was dispatched fourteen days after completion. The platform migrated regions in six waves between weeks 3 and 19 of a 22-week window, for reasons it recorded as operational. Wave assignment was not randomised, which motivates the estimators below.

Our analysis window covers 38,714 completed jobs and 2,146 active tradespeople. Table 1 reports the two strata.

	Immediate	14-day
Completed jobs	21,406	17,308
Active tradespeople	2,146	2,011
Submitted ratings	14,642	7,131
Response rate	68.4%	41.2%
Mean rating	4.71	4.12
90-day callback rate	11.3%	11.5%

Table 1: Analysis strata. Response rate is submitted ratings over completed jobs; callback rate is the platform’s ninety-day same-address, same-category flag.

3.2. Outcomes

Two outcomes are pre-specified. The first is the submitted rating on the platform’s five-point

scale, at job level. The second is the platform's own repeat-callback flag: a second booking at the same service address, in the same fault category, within ninety days of a completed job. The flag is generated by the platform's dispatch system for its own scheduling purposes and is not visible to households, tradespeople, or the ranking surface, which makes it the closest thing in the record to an outcome measured independently of the instrument under study.

3.3. Estimation

For job i in region r and week t we estimate

$$y_{irt} = \beta D_{rt} + \alpha_r + \gamma_t + x_{irt}^\top \delta + \varepsilon_{irt}$$

where D_{rt} indicates that region r had switched to the delayed prompt by week t , α_r and γ_t are region and week fixed effects, and x_{irt} carries trade category, job value band, and time of day. Because adoption is staggered and the treatment effect need not be homogeneous across waves, we report the two-way fixed-effects estimate alongside the heterogeneity-robust estimators of [19] and [20], and we follow [21] and [22] in checking the weights the two-way specification places on already-treated comparisons. Standard errors are clustered at the region level following [23], with the small-cluster caution of [24] noted: nine clusters is few, and we report wild-bootstrap intervals throughout.

4. Results

4.1. Ratings move

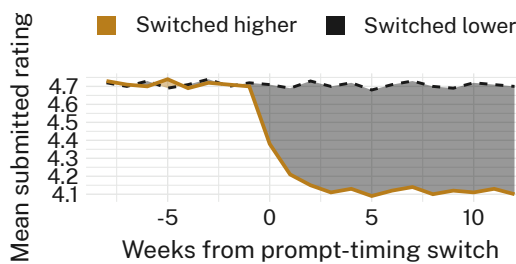


Figure 1: Weekly mean submitted rating in switched regions (gold) against regions not yet switched (dashed), indexed to the switch week. The band opens at week 0 and has not closed by week 12.

Submitted ratings fall on contact with the delayed prompt. Figure 1 plots the weekly mean in switched regions against regions not yet switched, indexed to each region's own switch week; the two series are indistinguishable for the eight weeks preceding the switch and sep-

arate within a single week of it. The difference-in-differences estimate is -0.55 points (95% CI -0.63 to -0.47). Estimates from [19] and [20] agree to within 0.03 points. We find no pre-trend in the eight weeks before each region's switch.

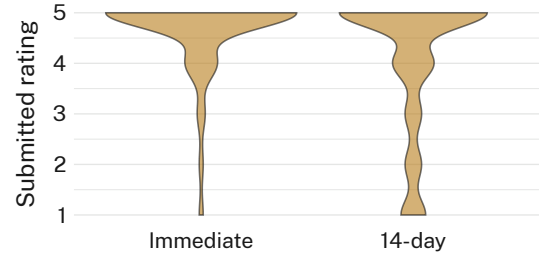


Figure 2: Distribution of submitted ratings by prompt timing. The delayed prompt is bimodal: the five-star share falls from 80.0% to 60.0% while the one-star share rises from 1.7% to 10.0%.

The mean understates what happens to the distribution. Figure 2 shows that the delayed prompt does not shift a single mass downward; it splits one. The five-star share falls from 80.0% to 60.0% and the one-star share rises from 1.7% to 10.0%, with the intermediate categories largely unchanged. Response rate falls from 68.4% to 41.2%, and the composition of the non-response changes with it: under the immediate prompt, non-response is concentrated in jobs completed outside business hours, while under the delayed prompt it is distributed evenly across the day.

4.2. The work does not

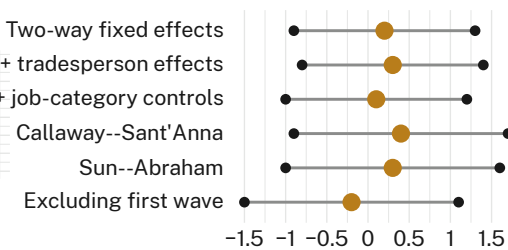


Figure 3: Callback-rate difference-in-differences under six specifications. Every interval spans zero and the point estimates span 0.6 percentage points.

The platform's recorded ninety-day callback rate is 11.3% before the switch and 11.5% after. The difference-in-differences estimate is $+0.2$ percentage points (95% CI -0.9 to $+1.3$). Figure 3 reports the same estimate under six specifications, including tradesperson fixed effects,

job-category controls, both heterogeneity-robust estimators, and a re-estimation dropping the first wave; the point estimates span 0.6 percentage points and every interval contains zero. Adding tradesperson fixed effects moves the estimate by 0.1 percentage points (n.s.); we retain the specification for completeness.

4.3. Only the delayed score knows anything

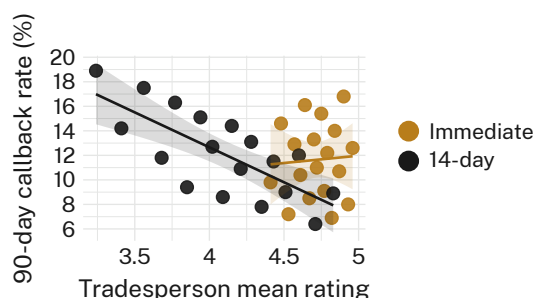


Figure 4: Tradesperson mean rating against recorded ninety-day callback rate. The fitted slope is flat under the immediate prompt and negative under the delayed one.

The two instruments differ in what they are associated with. Figure 4 plots each tradesperson’s mean rating against their recorded callback rate. Under the immediate prompt the association is absent ($r = 0.04$, 95% CI -0.05 to 0.13); under the delayed prompt it is present and in the expected direction ($r = -0.31$, 95% CI -0.39 to -0.23). The platform’s published ranking percentile, which weights the rating mean heavily, is likewise uncorrelated with callback rate over the immediate-prompt period ($r = 0.06$).

5. Discussion

The finding we consider most durable is the one that does not move. A change that shifts a headline score by more than half a point on a five-point scale, and reshapes its distribution, leaves the platform’s own record of whether the repair held exactly where it was. The score and the outcome were never coupled; the switch merely made that visible.

The correlational result suggests the coupling is recoverable, if modestly. A rating submitted after a fortnight of ordinary use has had the opportunity to observe something, and the association we find with callbacks indicates it sometimes does. Whether a platform wants that information at the price of twenty-seven percentage points

of response rate is a question of what the instrument is for, and we observe that the instrument is currently for ranking.

We read the arrangement as a governance property rather than a measurement error. Nothing in the immediate prompt is mis-specified; it asks a clear question at a well-defined moment, and it is answered promptly and in good faith. It is simply asked before the answer exists, and the ranking surface has been trained on the result in most settings we examined.

6. Limitations

Our evidence comes from one platform, though the consistency of the effect across all nine regions and all five trades suggests the mechanism is not platform-specific. Nine clusters is a small number for the inference we perform, which is why we report wild-bootstrap intervals; these are wider than the conventional ones and the ratings result survives them comfortably. The callback flag captures faults that return to the same platform, and would miss a household that gave up and called someone else — an omission that can only have understated the callback rate we already find unmoved.

7. Conclusion

A national home-repair platform moved its satisfaction prompt from the moment of completion to fourteen days afterwards, region by region, and we used the staggering to separate what the prompt measures from what the work achieves. Ratings fell by 0.55 points and split into a bimodal distribution; the platform’s own ninety-day callback record did not move by a detectable margin under any of six specifications. Only the delayed rating is associated with that record. The immediate prompt remains the default across the sector, and remains the input to the ranking surface. Future work might examine what a ranking surface fitted to the delayed instrument would reorder, and over what horizon.

References

- [1] R. A. Peterson and W. R. Wilson, “Measuring Customer Satisfaction: Fact and Artifact,” *Journal of the Academy of Marketing Science*, vol. 20, no. 1, pp. 61–71, 1992, doi: [10.1007/BF02723476](https://doi.org/10.1007/BF02723476).
- [2] J. J. Cronin and S. A. Taylor, “Measuring Service Quality: A Reexamination

- and Extension,” *Journal of Marketing*, vol. 56, no. 3, pp. 55–68, 1992, doi: [10.1177/002224299205600304](https://doi.org/10.1177/002224299205600304).
- [3] E. W. Anderson and M. W. Sullivan, “The Antecedents and Consequences of Customer Satisfaction for Firms,” *Marketing Science*, vol. 12, no. 2, pp. 125–143, 1993, doi: [10.1287/mksc.12.2.125](https://doi.org/10.1287/mksc.12.2.125).
- [4] D. A. Redelmeier and D. Kahneman, “Patients’ Memories of Painful Medical Treatments: Real-Time and Retrospective Evaluations of Two Minimally Invasive Procedures,” *Pain*, vol. 66, no. 1, pp. 3–8, 1996, doi: [10.1016/0304-3959\(96\)02994-6](https://doi.org/10.1016/0304-3959(96)02994-6).
- [5] D. Kahneman, B. L. Fredrickson, C. A. Schreiber, and D. A. Redelmeier, “When More Pain Is Preferred to Less: Adding a Better End,” *Psychological Science*, vol. 4, no. 6, pp. 401–405, 1993, doi: [10.1111/j.1467-9280.1993.tb00589.x](https://doi.org/10.1111/j.1467-9280.1993.tb00589.x).
- [6] N. M. Bradburn, L. J. Rips, and S. K. Shevell, “Answering Autobiographical Questions: The Impact of Memory and Inference on Surveys,” *Science*, vol. 236, no. 4798, pp. 157–161, 1987, doi: [10.1126/science.3563494](https://doi.org/10.1126/science.3563494).
- [7] J. J. Fenton, A. F. Jerant, K. D. Bertakis, and P. Franks, “The Cost of Satisfaction: A National Study of Patient Satisfaction, Health Care Utilization, Expenditures, and Mortality,” *Archives of Internal Medicine*, vol. 172, no. 5, pp. 405–411, 2012, doi: [10.1001/archinternmed.2011.1662](https://doi.org/10.1001/archinternmed.2011.1662).
- [8] M. P. Manary, W. Boulding, R. Staelin, and S. W. Glickman, “The Patient Experience and Health Outcomes,” *New England Journal of Medicine*, vol. 368, no. 3, pp. 201–203, 2013, doi: [10.1056/NEJMp1211775](https://doi.org/10.1056/NEJMp1211775).
- [9] J. Sitzia and N. Wood, “Patient Satisfaction: A Review of Issues and Concepts,” *Social Science & Medicine*, vol. 45, no. 12, pp. 1829–1843, 1997, doi: [10.1016/S0277-9536\(97\)00128-7](https://doi.org/10.1016/S0277-9536(97)00128-7).
- [10] C. Dellarocas, “The Digitization of Word of Mouth: Promise and Challenges of Online Feedback Mechanisms,” *Management Science*, vol. 49, no. 10, pp. 1407–1424, 2003, doi: [10.1287/mnsc.49.10.1407.17308](https://doi.org/10.1287/mnsc.49.10.1407.17308).
- [11] S. Tadelis, “Reputation and Feedback Systems in Online Platform Markets,” *Annual Review of Economics*, vol. 8, no. 1, pp. 321–340, 2016, doi: [10.1146/annurev-economics-080315-015325](https://doi.org/10.1146/annurev-economics-080315-015325).
- [12] G. Zervas, D. Proserpio, and J. W. Byers, “A First Look at Online Reputation on Airbnb, Where Every Stay Is Above Average,” *Marketing Letters*, vol. 32, no. 1, pp. 1–16, 2020, doi: [10.1007/s11002-020-09546-4](https://doi.org/10.1007/s11002-020-09546-4).
- [13] C. Nosko and S. Tadelis, “The Limits of Reputation in Platform Markets: An Empirical Analysis and Field Experiment,” *NBER Working Paper 20830*, National Bureau of Economic Research, 2015, doi: [10.3386/w20830](https://doi.org/10.3386/w20830).
- [14] L. Muchnik, S. Aral, and S. J. Taylor, “Social Influence Bias: A Randomized Experiment,” *Science*, vol. 341, no. 6146, pp. 647–651, 2013, doi: [10.1126/science.1240466](https://doi.org/10.1126/science.1240466).
- [15] T. L. Keiningham, B. Cooil, T. W. Andreassen, and L. Aksoy, “A Longitudinal Examination of Net Promoter and Firm Revenue Growth,” *Journal of Marketing*, vol. 71, no. 3, pp. 39–51, 2007, doi: [10.1509/jmkg.71.3.39](https://doi.org/10.1509/jmkg.71.3.39).
- [16] R. M. Groves and E. Peytcheva, “The Impact of Nonresponse Rates on Nonresponse Bias: A Meta-Analysis,” *Public Opinion Quarterly*, vol. 72, no. 2, pp. 167–189, 2008, doi: [10.1093/poq/nfn011](https://doi.org/10.1093/poq/nfn011).
- [17] M. Hanke and T. Solberg, “Nobody’s Unhappy at 6 AM: Timestamp-Clustering a Restroom Satisfaction Kiosk’s Button Presses Against the Cleaning Roster,” *School of Continuous Improvement, Slop University*, 2026, doi: [10.5555/slop.1rx612](https://doi.org/10.5555/slop.1rx612).
- [18] L. Achterberg and F. Okoro, “Still Under Review: A Field Study of Portal Status Displays and Applicant Persistence After an Internal-Transfer Process Goes Quiet,” *School of Emergent Priorities, Slop University*, 2026, doi: [10.5555/slop.kbdcdq](https://doi.org/10.5555/slop.kbdcdq).
- [19] B. Callaway and P. H. C. Sant’Anna, “Difference-in-Differences with Multiple Time Periods,” *Journal of Econometrics*, vol. 225, no. 2, pp. 200–230, 2021, doi: [10.1016/j.jeconom.2020.12.001](https://doi.org/10.1016/j.jeconom.2020.12.001).
- [20] L. Sun and S. Abraham, “Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects,” *Journal of Econometrics*, vol. 225,

- no. 2, pp. 175–199, 2021, doi: [10.1016/j.jeconom.2020.09.006](https://doi.org/10.1016/j.jeconom.2020.09.006).
- [21] A. Goodman-Bacon, “Difference-in-Differences with Variation in Treatment Timing,” *Journal of Econometrics*, vol. 225, no. 2, pp. 254–277, 2021, doi: [10.1016/j.jeconom.2021.03.014](https://doi.org/10.1016/j.jeconom.2021.03.014).
- [22] C. de Chaisemartin and X. D'Haultfœuille, “Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects,” *American Economic Review*, vol. 110, no. 9, pp. 2964–2996, 2020, doi: [10.1257/aer.20181169](https://doi.org/10.1257/aer.20181169).
- [23] A. C. Cameron and D. L. Miller, “A Practitioner's Guide to Cluster-Robust Inference,” *Journal of Human Resources*, vol. 50, no. 2, pp. 317–372, 2015, doi: [10.3368/jhr.50.2.317](https://doi.org/10.3368/jhr.50.2.317).
- [24] M. Bertrand, E. Duflo, and S. Mullainathan, “How Much Should We Trust Differences-in-Differences Estimates?,” *The Quarterly Journal of Economics*, vol. 119, no. 1, pp. 249–275, 2004, doi: [10.1162/003355304772839588](https://doi.org/10.1162/003355304772839588).